

CBE 3610 Handbook

August 20, 2026

All information in this Reference Book is adapted from or inspired by the following source except where indicated otherwise:

Fogler, H. S., *Elements of Chemical Reaction Engineering*, 5th Edition, Pearson Education, Inc., Boston, 2016, ISBN: 978-0-13-388751-8.

1 Purpose of This Handbook:

This course is built around one central idea: you learn chemical reaction engineering by *engaging with it*, not by memorizing equations. The goal of the CBE 3610 Handbook is therefore not to replace understanding. Rather, its purpose is to help you focus on the parts of reactor design that matter most: identifying the governing principles, choosing an appropriate approach, making reasonable assumptions, carrying out the analysis, and interpreting the results.

1. I do not expect you to memorize every equation used in chemical reaction engineering. Research consistently shows that learning is deeper when students focus on concepts, problem-solving, and engineering reasoning rather than rote memorization. Consequently, this Handbook contains the core equations, definitions, variable descriptions, and relationships that we use throughout the course. Rather than memorizing formulas, I encourage you to learn where equations come from, when they apply, what assumptions they require, and how they connect to the governing balances.
2. In chemical reaction engineering, knowing *how to start* is often more important than remembering a specific equation. When students become stuck, the solution is usually to identify the reactor type, determine whether equilibrium matters, determine whether temperature effects matter, and return to the governing mole and energy balances. This Handbook is designed to help guide that process by collecting the most commonly used equations and concepts in one place.
3. I also believe strongly in affordable and accessible learning. The course textbook is available to Ohio State students at no cost through the university libraries, and purchasing a textbook is

not required. As a result, I wanted to provide a resource that contains the fundamental information needed to succeed in the course regardless of whether you have the textbook open in front of you. In addition to equations, this Handbook includes definitions, explanations, units, and clarifications that I hope make the material more approachable.

4. Finally, this Handbook is intended to support the development of engineering judgment. Real reactor design problems rarely begin with an obvious path forward. The challenge is not simply solving equations; it is choosing an appropriate model, understanding its assumptions and limitations, and deciding whether the result is physically reasonable. I hope this Handbook helps you build that confidence throughout the semester.

When you are unsure where to start, open this book, identify the governing balances, and take the next reasonable step. You do not need to understand everything immediately. If you engage consistently, ask questions, and keep moving forward, you can succeed in chemical reaction engineering.

Happy reactor designing!

— RBG

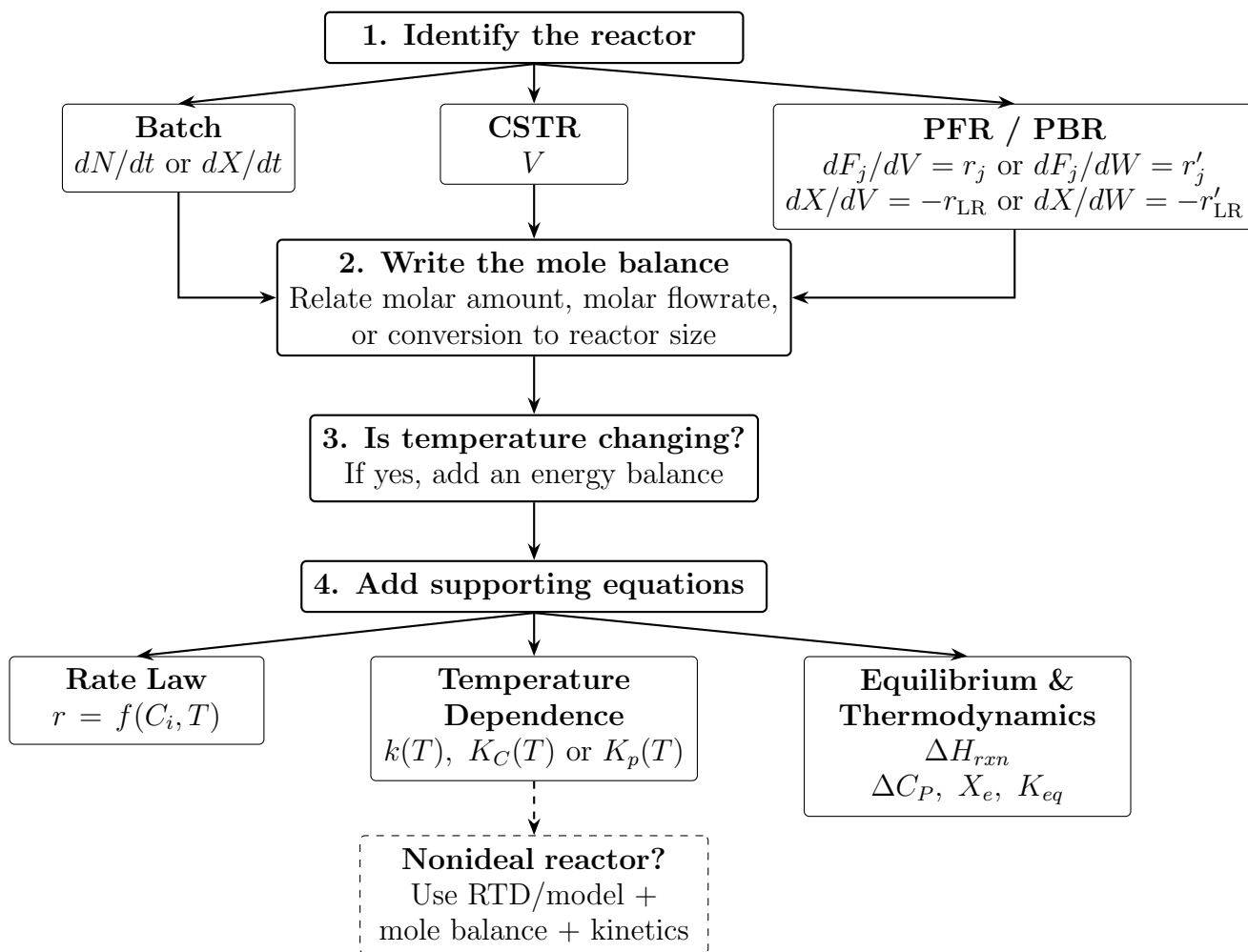
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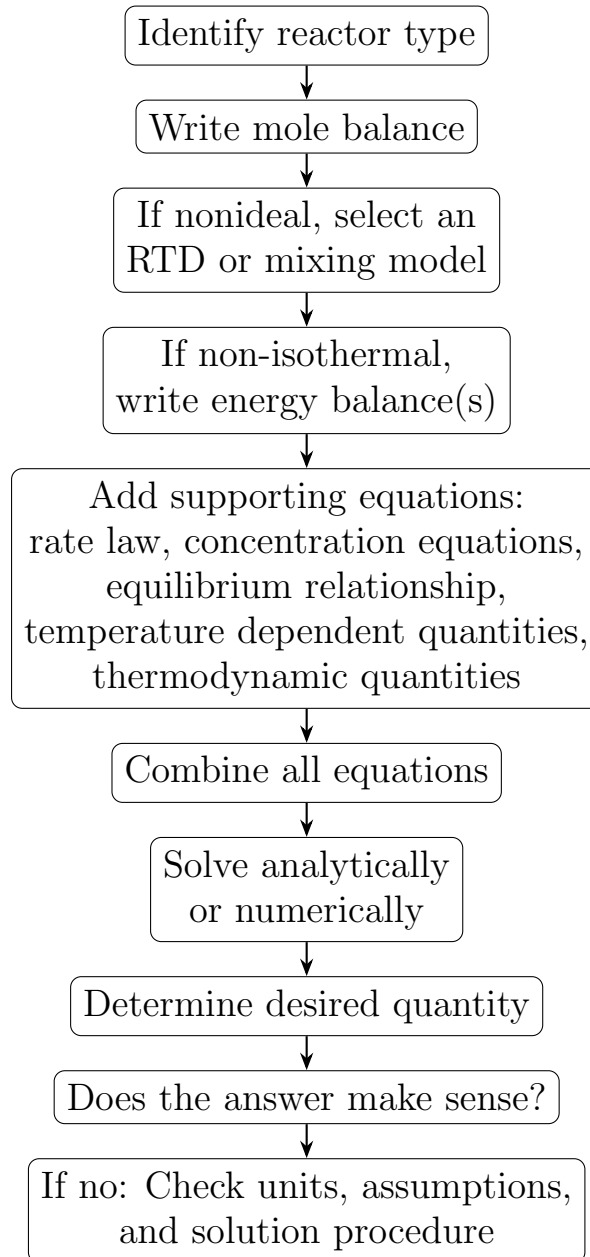
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2 How Do I Start?



3 How Do I Solve CRE Problems?



4 Definitions/Descriptions of Useful CRE Quantities and Terms

4.1 Gas Constant Values

$$R = \frac{0.73 \text{ ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot ^\circ\text{R}} \quad R = \frac{1.987 \text{ Btu}}{\text{lbmol} \cdot ^\circ\text{R}}$$

$$R = \frac{8.314 \text{ dm}^3 \cdot \text{kPa}}{\text{mol} \cdot \text{K}} \quad R = \frac{8.314 \text{ J}}{\text{mol} \cdot \text{K}}$$

$$R = \frac{0.082 \text{ dm}^3 \cdot \text{atm}}{\text{mol} \cdot \text{K}} \quad R = \frac{1.987 \text{ cal}}{\text{mol} \cdot \text{K}}$$

Boltzmann's constant: $k_B = 1.381 \times 10^{-23} \frac{\text{J}}{\text{molecule} \cdot \text{K}}$

4.2 Other Useful Definitions and Descriptions

The limiting reactant is often denoted “LR”.

The conversion of B, $X_B = \frac{\text{Moles of B reacted}}{\text{Moles of B fed or present initially}}$. X is often written with respect to the limiting reactant; however, it can be calculated for any reactant.

For flow reactors, the space time $\frac{\text{system volume}}{\text{inlet volumetric flowrate}} = \frac{V}{v_0}$. In an ideal CSTR or PFR, space time equals residence time, τ .

The change in the total number of moles per mole of limiting reactant (LR) reacted, $\delta = \frac{\text{Change in the total number of moles}}{\text{Mole of LR reacted}}$.

Ratio of molar amounts or molar flowrates, $\Theta_B = \frac{\text{Moles of species B fed or present initially}}{\text{Moles of LR fed or present initially}}$.

For gas phase systems, ϵ quantifies a change in the total number of moles: $\epsilon = \frac{\text{Change in total number of moles for complete conversion}}{\text{Total moles fed}} = y_{\text{LR},0}\delta$.

The Damköhler number, Da , is the ratio of the characteristic reaction rate to the characteristic transport (convection) rate. For a simple

power law reaction, $Da = kC_{A0}^{n-1}\tau$.

The instantaneous selectivity, $S_{D/U} = \frac{r_D}{r_U}$, is written to quantify the ratio of the rate of formation of the desired product to that of the undesired product.

The overall selectivity, $\tilde{S}_{D/U} = \frac{F_{D|exit}}{F_{U|exit}}$, is written to quantify the ratio of the molar flowrate of the desired product to that of the undesired product.

The instantaneous yield, $Y_D = \frac{r_D}{-r_{LR}}$, is written to quantify the ratio of the rate of formation of the desired product to the rate of consumption of the limiting reactant.

The overall yield, $\tilde{Y}_D = \frac{\text{Moles of desired product formed}}{\text{Moles of LR reacted}}$.

The definition of *parallel* is: Processes that occur simultaneously and independently. The definition of *serial* is: Processes that occur sequentially, where each individual event depends in at least some way on the events prior.

An *isothermal process* maintains constant temperature. An *isobaric process* maintains constant pressure. An *isochoric process* maintains constant volume. An *adiabatic process* is one in which there is no heat transfer between the system and its surroundings.

The *heat generated* term, Q_g or $G(T)$, is due to the reaction.

The *heat removed* term, Q_r or $R(T)$, is due to 1) the heat exchanger, and 2) the sensible and/or latent heat changes that bring the feed to the reaction temperature.

In the Fogler textbook, $G(T)$ and $R(T)$ are normalized heat generated and heat removed curves used in the analysis of non-isothermal CSTRs. In contrast, Q_g and Q_r typically refer to the rates of heat generation and heat removal.

The *ignition temperature* for a tank reactor is the feed temperature at which a fraction of a degree increase would cause a jump in the reactor temperature from the lower steady state temperature to the upper steady state temperature. The *extinction temperature* is the feed temperature at which a fraction of a degree decrease would cause a jump in the reactor temperature from the upper steady state temperature to the lower steady state temperature.

Quantities with a dot over them are time derivatives or flow rates. E.g., $\dot{Q}$ is heat flow per unit time, and has units of, e.g., J/s.

Quantities with a “0” subscript are for inlet or initial conditions, depending on reactor type. For flow reactors, the 0 subscript indicates inlet conditions. For batch reactors, the 0 subscript indicates initial conditions.

5 Heat of Reaction and Heat Capacity

5.1 Units

The following formulas require the heat of reaction, ΔH_{rxn}° . The units of ΔH_{rxn}° are typically expressed as energy per mole. The mole basis must be specified. Common choices include kJ/mol A reacted, kJ/mol

B reacted, or kJ/mol C formed. For example, consider the reaction



If

$$\Delta H_{rxn}^{\circ} = -100 \text{ kJ/mol A reacted,}$$

then the total energy released by the reaction does not change. Only the normalization basis changes. The heat of reaction can also be expressed as

$$-50 \text{ kJ/mol B reacted}$$

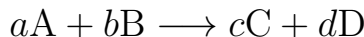
or

$$-25 \text{ kJ/mol C formed.}$$

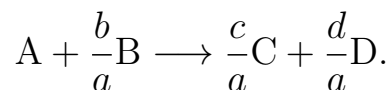
When using the energy balance, the mole units associated with ΔH_{rxn}° must be consistent with those used in the reaction-rate term because these quantities are multiplied together. For example, if the mole balance and rate law are written in terms of species A, then the rate will typically be expressed as r_A , which has units of moles of A per volume per time. In this case, ΔH_{rxn}° should be expressed in units of energy per mole of A:

$$(-r_A)(-\Delta H_{rxn,A}) = \left(\frac{\text{mol A}}{\text{volume} \cdot \text{time}} \right) \left(\frac{\text{kJ}}{\text{mol A}} \right) = \frac{\text{kJ}}{\text{volume} \cdot \text{time}}$$

A convenient way to obtain ΔH_{rxn}° in the proper units is to rewrite the reaction so that the stoichiometric coefficient of the species appearing in the rate expression is equal to one. For example, if the rate is written in terms of species A, then rewrite



as



The resulting value of ΔH_{rxn}° will then have units of energy per mole of A reacted.

$$\Delta H_{rxn,A}^\circ = \left(\frac{d}{a} H_D^\circ + \frac{c}{a} H_C^\circ - \frac{b}{a} H_B^\circ - H_A^\circ \right) \quad (1)$$

where the H° are heats of formation measured at the reference temperature T_{ref} . The “A” subscript indicates that species A is taken as the basis. Calculated this way, $\Delta H_{rxn,A}^\circ$ has units of energy/mole A (e.g., kJ/mol A).

Similarly,

$$\Delta C_{P,A} = \left(\frac{d}{a} C_{P,D} + \frac{c}{a} C_{P,C} - \frac{b}{a} C_{P,B} - C_{P,A} \right) \quad (2)$$

gives the change in heat capacity per mole A reacted. ΔC_P can be positive, negative, or zero depending on the stoichiometry and species heat capacities.

5.2 As Functions of T

Because reaction enthalpies are calculated from species enthalpies, we can determine how ΔH_{rxn} changes with temperature by first considering how the individual species enthalpies (H_j) change with temperature.

H are related to C_P by

$$H(T) - H(T_{ref}) = \int_{T_{ref}}^T C_P(T) dT \quad (3)$$

If we can assume that C_P is constant (i.e., that it does not depend on T), then

$$H(T) = H(T_{ref}) + C_P(T - T_{ref}) \quad (4)$$

The approximation that C_P is constant is usually reasonable over moderate temperature ranges but may become inaccurate over very large temperature changes.

Combining Equations 1, 2, and 4

$$\Delta H_{rxn}(T) = \Delta H_{rxn}^\circ(T_{ref}) + \Delta C_P(T - T_{ref}) \quad (5)$$

In Equation 5, $\Delta C_P > 0$ means that the reaction becomes less negative (more positive) as temperature increases, whereas $\Delta C_P < 0$ means that the reaction becomes more negative (less positive) as temperature increases.

6 Levenspiel Plot Equations

A Levenspiel plot is a graph of $\frac{F_{LR,0}}{-r_{LR}}$ vs. conversion X , where X is defined with respect to the limiting reactant (LR).

For a CSTR:

$$V = \left(\frac{F_{LR,0}}{-r_{LR}(X)} \right) (X_{out} - X_{in}) \quad (6)$$

In other words, the volume of a CSTR is equal to the area of a rectangle on the Levenspiel plot, where

$$\text{height} = \frac{F_{LR,0}}{-r_{LR}}$$

and

$$\text{width} = X_{out} - X_{in}$$

For a PFR:

$$V = F_{LR,0} \int_{X_{in}}^{X_{out}} \frac{dX}{-r_{LR}(X)} \quad (7)$$

In other words, the volume of a PFR is equal to the area under the Levenspiel plot between the inlet and outlet conversions.

7 Common Reactors

Assumptions Used in This Section

Unless otherwise stated, the equations in this section assume:

- Steady-state operation for flow reactors
- Perfect mixing in CSTRs
- Plug flow in PFRs and PBRs
- Negligible kinetic and potential energy effects
- Constant heat-transfer coefficients
- Shaft work can be neglected (may not be true for viscous fluids)
- A heat exchanger with constant T_a (for other heat exchange scenarios, see Section 12).

Important

Whenever a rate term is multiplied by a heat-of-reaction term, both quantities must be written on the same stoichiometric basis. Examples:

$$r_A \leftrightarrow \Delta H_{rxn,A} \leftrightarrow \Delta C_{P,A}$$

$$r_B \leftrightarrow \Delta H_{rxn,B} \leftrightarrow \Delta C_{P,B}$$

If the balance equations are written in terms of conversion X , then X , r_{LR} , $\Delta H_{rxn,LR}$, and $\Delta C_{P,LR}$ must be written with respect to the limiting reactant.

Note: The balances in this section are written for species A when writing balances for a specific species. The subscript LR is used when writing balances in terms of conversion.

7.1 Continuously stirred tank reactor (CSTR)

7.1.1 Mole Balance

In terms of F :

$$V = \frac{F_{A0} - F_A}{(-r_A)_{out}} \quad (8)$$

where the subscript *out* indicates that the rate must be evaluated at the exit conditions since CSTRs are perfectly mixed.

In terms of X :

$$V = \frac{F_{LR,0} (X_{out} - X_{in})}{(-r_{LR})_{out}} \quad (9)$$

This equation can be used to solve for X . The resulting equation is sometimes denoted X_{MB} .

7.1.2 Energy Balance

In terms of F :

$$V = \frac{UA(T_a - T) + \sum_j F_{j0} C_{Pj} (T_0 - T)}{(-r_B) (-\Delta H_{rxn,B}(T))} \quad (10)$$

Multiple reactions (q reactions and m species):

$$V = \frac{UA(T_a - T) + \sum_j^m F_{j0} C_{Pj} (T_0 - T)}{\sum_i^q (-r_{iB}) (-\Delta H_{rxn,iB}(T))} \quad (11)$$

In terms of X :

$$X_{EB} = \frac{\left(\frac{UA}{F_{LR,0}} (T_a - T) \right) + \sum_j \Theta_j C_{Pj} (T_0 - T)}{\Delta H_{rxn,LR}(T)} \quad (12)$$

Notes:

1. This form is written using the inlet heat capacity contributions, $\sum_j \Theta_j C_{Pj} (T_0 - T)$. Alternative energy balance formulations may instead be written using ΔC_P and yield terms such as $\sum_j \Theta_j C_{Pj} (T_0 - T) + \Delta C_P X$. Both approaches are equivalent *when derived consistently*.
2. Here the $\Delta H_{rxn,LR}(T)$ term includes the temperature correction.

In the Fogler heat generated/heat removed graphical method, $G(T)$ and $R(T)$ are obtained by multiplying X_{MB} and X_{EB} , respectively, by $(-\Delta H_{rxn})$.

7.2 Plug flow reactor (PFR)

7.2.1 Mole Balance

In terms of F :

$$\frac{dF_A}{dV} = r_A \quad (13)$$

In terms of X :

$$F_{LR,0} \frac{dX_{LR}}{dV} = -r_{LR} \quad (14)$$

7.2.2 Energy Balance

In terms of F :

$$\frac{dT}{dV} = \frac{(-r_B) (-\Delta H_{rxn,B}(T)) + Ua (T_a - T)}{\sum_j F_j C_{Pj}} \quad (15)$$

Notes: The heat capacity term is evaluated using local conditions and therefore varies with reactor position. $a = dA/dV$ is the heat transfer area per reactor volume.

Multiple reactions (q reactions and m species):

$$\frac{dT}{dV} = \frac{\sum_i^q (-r_{iB}) (-\Delta H_{rxn,iB}(T)) + Ua (T_a - T)}{\sum_j^m F_j C_{Pj}} \quad (16)$$

In terms of X :

$$\frac{dT}{dV} = \frac{(-r_{LR}) (-\Delta H_{rxn}(T)) + Ua (T_a - T)}{F_{LR,0} \left(\sum_j \Theta_j C_{Pj} + \Delta C_P X \right)} \quad (17)$$

7.3 Batch Reactor

7.3.1 Mole Balance

in terms of N :

$$\frac{dN_A}{dt} = r_A V \quad (18)$$

in terms of X :

$$N_{LR,0} \frac{dX_{LR}}{dt} = -r_{LR} V \quad (19)$$

7.3.2 Energy Balance

in terms of N , single reaction:

$$\frac{dT}{dt} = \frac{(-r_B V) (-\Delta H_{rxn,B}(T)) + U A (T_a - T)}{\sum_j N_j C_{Pj}} \quad (20)$$

Multiple reactions (q reactions and m species):

$$\frac{dT}{dt} = \frac{V \sum_{i=1}^q (-r_{iB}) (-\Delta H_{rxn,iB}(T)) + U A (T_a - T)}{\sum_{j=1}^m N_j C_{Pj}} \quad (21)$$

in terms of X :

$$\frac{dT}{dt} = \frac{V (-r_{LR}) (-\Delta H_{rxn}(T)) + U A (T_a - T)}{N_{LR,0} \left(\sum_j \Theta_j C_{Pj} + \Delta C_P X \right)} \quad (22)$$

7.4 Packed Bed Reactor (PBR)

7.4.1 Mole Balance

in terms of F :

$$\frac{dF_A}{dW} = r'_A \quad (23)$$

in terms of X :

$$F_{LR0} \frac{dX_{LR}}{dW} = -r'_{LR} \quad (24)$$

7.4.2 Energy Balance

in terms of F :

$$\frac{dT}{dW} = \frac{(-r'_B)(-\Delta H_{rxn,B}(T)) + \frac{U_a}{\rho_b}(T_a - T)}{\sum_j F_j C_{Pj}} \quad (25)$$

Notes: The heat capacity term is evaluated using local conditions and therefore varies with reactor position. Here $a = dA/dV$ is the heat transfer area per reactor volume. ρ_b is the heat transfer coefficient per mass catalyst.

Multiple reactions (q reactions and m species):

$$\frac{dT}{dW} = \frac{\sum_i^q (-r'_{iB})(-\Delta H_{rxn,iB}(T)) + \frac{U_a}{\rho_b}(T_a - T)}{\sum_j^m F_j C_{Pj}} \quad (26)$$

in terms of X :

$$\frac{dT}{dW} = \frac{(-r'_{LR})(-\Delta H_{rxn}(T)) + \frac{U_a}{\rho_b}(T_a - T)}{F_{LR,0} \left(\sum_j \Theta_j C_{Pj} + \Delta C_P X \right)} \quad (27)$$

8 General Form of the Rate Law with Examples

Rate of consumption of species A:

$$-r_A = [k_A(T)] [f(C_A, C_B, \dots)] \quad (28)$$

E.g., f could follow a power law model, e.g.,

$$f(C_A, C_B, \dots) = C_A^\alpha C_B^\beta \dots \quad (29)$$

Note: In general, reaction orders ($\alpha, \beta, \dots$) are determined experimentally and are not necessarily equal to the stoichiometric coefficients. For an elementary reaction, the reaction orders with respect to the reactants equal their stoichiometric coefficients.

For a single reactant:

$$-r_A = k C_A^n \quad (30)$$

The units on k depend on the overall reaction order.

For two reactants:

$$-r_A = k C_A^n C_B^m \quad (31)$$

A common form of a reversible rate law is:

$$-r_A = k_f \left(\prod_{\text{reactants}} C_i^{|\nu_i|} - \frac{\prod_{\text{products}} C_j^{|\nu_j|}}{K_C} \right) \quad (32)$$

where k_f is the forward rate constant, defined in terms of one of the reactant species, K_C is the equilibrium constant with respect to concentration, and ν are stoichiometric coefficients. We take the absolute

values of ν (which are negative for reactants and positive for products). The stoichiometry used to define k_f , K_C , and the rate expression must be consistent.

Example reversible reaction, $aA + bB \rightleftharpoons cC + dD$:

$$-r_A = k_{f,A} \left(C_A^a C_B^b - \frac{C_C^c C_D^d}{K_C} \right) \quad (33)$$

where A is used to define the stoichiometry of K_C and k_f .

Relative rates: For the reaction $aA + bB \rightarrow cC + dD$,

$$\frac{-r_A}{a} = \frac{-r_B}{b} = \frac{r_C}{c} = \frac{r_D}{d} \quad (34)$$

This relationship allows rates written with respect to one species to be converted into rates written with respect to another species.

9 Equilibrium Relationships

For the reaction $A + \frac{b}{a}B \rightleftharpoons \frac{c}{a}C + \frac{d}{a}D$, the equilibrium constant in terms of pressure,

$$K_p = \frac{P_{C,\text{eq}}^{c/a} P_{D,\text{eq}}^{d/a}}{P_{A,\text{eq}} P_{B,\text{eq}}^{b/a}} \quad (35)$$

The equilibrium constant in terms of concentration:

$$K_C = \frac{C_{C,\text{eq}}^{c/a} C_{D,\text{eq}}^{d/a}}{C_{A,\text{eq}} C_{B,\text{eq}}^{b/a}} \quad (36)$$

In Equations 35 and 36, the subscript “eq” denotes that the partial pressures or concentrations must be at their equilibrium values.

For an ideal gas:

$$K_{eq} = \frac{K_p}{(1 \text{ bar})^\delta} \quad (37)$$

For an ideal liquid:

$$K_{eq} = \frac{K_C}{(1 \text{ M})^\delta} \quad (38)$$

where K_{eq} is defined as

$$K_{eq} = \prod_j a_j^{\nu_j} \quad (39)$$

where

$$a_j = \frac{P_j}{P^\circ} \quad (40)$$

for ideal gases and

$$a_j = \frac{C_j}{C^\circ} \quad (41)$$

for ideal solutions. These factors nondimensionalize the equilibrium constant so that K_{eq} may be used in thermodynamic relationships such as

$$K_{eq} = \exp\left(\frac{-\Delta G_{rxn}^\circ}{RT}\right) \quad (42)$$

In this handbook:

- K_C is the equilibrium constant based on concentration

- K_p is the equilibrium constant based on pressure
- K_{eq} is the thermodynamic (dimensionless) equilibrium constant

These expressions can be used to solve for the partial pressures or concentrations at equilibrium.

From thermodynamics,

$$\Delta G_{rxn} = \Delta H_{rxn} - T \Delta S_{rxn} \quad (43)$$

For an ideal gas:

$$K_p = K_c (RT)^\delta \quad (44)$$

Because liquid phase reactions are generally described using concentrations rather than partial pressures, K_C is typically used instead of K_p . Further, Equation 44 is derived from the ideal gas equation and hence is meaningless for a liquid.

The approach to equilibrium, β

$$\beta = \frac{Q_C}{K_C} = \frac{Q_p}{K_p} \quad (45)$$

where Q_C and Q_p are the reaction quotients:

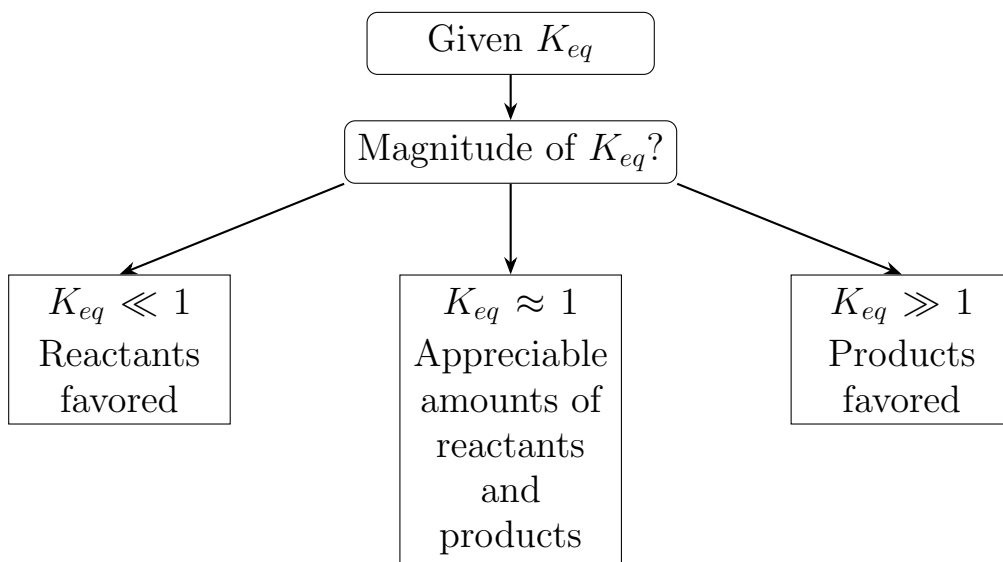
$$Q_C = \frac{C_C^{c/a} C_D^{d/a}}{C_A C_B^{b/a}} \quad (46)$$

and

$$Q_p = \frac{P_C^{c/a} P_D^{d/a}}{P_A P_B^{b/a}} \quad (47)$$

i.e., Q_C and Q_p involve the actual concentrations or partial pressures, while K_C and K_p involve the equilibrium values. If $Q < K$, the reaction proceeds in forward. If $Q > K$, the reaction proceeds in reverse. If $Q = K$, the reaction is at equilibrium. Also at equilibrium, $\beta = 1$.

Using the magnitude of K_{eq} to estimate equilibrium composition:



10 Temperature dependent quantities

Specific rate constant for species A (Arrhenius Equation):

$$k_A(T) = A \exp\left(-\frac{\Delta E_{act}}{RT}\right) \quad (48)$$

where A is the pre-exponential term or frequency factor. In the Arrhenius form, increasing temperature generally increases the rate constant because a larger fraction of molecules have sufficient energy to overcome the activation barrier.

Temperature-dependence of rate constant (Arrhenius Equation):

$$k(T) = k(T_{ref}) \exp \left[\frac{\Delta E_{act}}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right] \quad (49)$$

Note that this equation assumes that A is constant.

Temperature-dependence of K_{eq} (van't Hoff equation):

$$K_{eq}(T) = K_{eq}(T_{ref}) \exp \left[\frac{\Delta H_{rxn}^{\circ}}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right] \quad (50)$$

Note that Equation 50 assumes that ΔH_{rxn}° is constant over the temperature range. Further, for endothermic reactions ($\Delta H_{rxn}^{\circ} > 0$), increasing temperature increases K_{eq} . For exothermic reactions ($\Delta H_{rxn}^{\circ} < 0$), increasing temperature decreases K_{eq} .

For gas phase reactions, the concentration equations also depend on T . See Section 11.

11 Concentration Equations

Concentration is Moles $\div$ Volume, i.e.,

$$C = \frac{N}{V} \quad (51)$$

in a batch system and

$$C = \frac{F}{v} \quad (52)$$

in a flow system. Note that $F = N/t$ and $v = V/t$.

Concentration has units of moles per volume. $1 \text{ M} = 1 \text{ mol/liter} = 1 \text{ mol/dm}^3 = 1 \text{ kmol/m}^3 = 0.062 \text{ lbmol/ft}^3$.

11.1 Liquid Phase

Volume and volumetric flowrate are assumed constant.

$$C_A = \frac{N_A}{V_0} \quad (53)$$

in a batch system and

$$C_A = \frac{F_A}{v_0} \quad (54)$$

in a flow system.

In terms of X :

$$C_B = C_{LR,0} \left(\Theta_B - \frac{\nu_B}{\nu_{LR}} X \right) \quad (55)$$

where X is defined with respect to the limiting reactant.

11.2 Gas Phase

All equations in this section assume ideal gas.

The concentration of species A,

$$C_A = \frac{P_A}{RT} = \frac{y_A P}{RT} \quad (56)$$

where P_A is the partial pressure of A, y_A is the mole fraction of A, and P is the total pressure.

Change in volumetric flowrate for ideal gases (general equation):

$$v = v_0 \left(\frac{F_T}{F_{T0}} \right) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (57)$$

Change in concentration for ideal gases (general equation):

$$C_B = C_{T0} \left(\frac{F_B}{F_T} \right) \left(\frac{P_T}{P_{T0}} \right) \left(\frac{T_0}{T} \right) \quad (58)$$

where C_{T0} is the total concentration in the feed (or present initially).
Change in partial pressure for ideal gases (general equation):

$$P_B = y_B P_T = y_B P_{T0} p \quad (59)$$

where $p = P_T/P_{T0}$. Note that temperature does not appear explicitly in the expression for P_B . Temperature influences concentration directly through the ideal gas law, whereas partial pressure depends on total pressure and composition.

Change in moles: For the reaction $aA + bB \longrightarrow cC + dD$, where A is the limiting reactant:

$$\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \quad (60)$$

δ must be written in terms of the limiting reactant.

Mole change parameter:

$$\epsilon = y_{LR,0} \delta \quad (61)$$

where $y_{LR,0}$ is the inlet/initial mole fraction of the limiting reactant.

Change in volumetric flowrate for ideal gases (for cases that can be written in terms of conversion, X):

$$v = v_0 (1 + \epsilon X) \left(\frac{P_0}{P} \right) \left(\frac{T}{T_0} \right) \quad (62)$$

Change in concentration for ideal gases (for cases that can be written in terms of conversion, X):

$$C_B = \frac{C_{LR,0} \left(\Theta_B - \left(\frac{\nu_B}{\nu_{LR}} \right) X \right)}{1 + \epsilon X} \left(\frac{P_T}{P_{T0}} \right) \left(\frac{T_0}{T} \right) \quad (63)$$

where ν_j are stoichiometric coefficients, which by definition are negative for reactants and positive for products.

Quick Guide

Liquid phase:

$$C_B = C_{LR,0} \left(\Theta_B - \frac{\nu_B}{\nu_{LR}} X \right) \text{ or } \frac{F_B \text{ or } N_B}{v_0 \text{ or } V_0}$$

Gas phase:

$$C_B = \frac{C_{LR,0} \left(\Theta_B - \frac{\nu_B}{\nu_{LR}} X \right)}{1 + \epsilon X} \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right) \text{ or } C_{T0} \left(\frac{F_B}{F_T} \right) \left(\frac{P_T}{P_{T0}} \right) \left(\frac{T_0}{T} \right)$$

where:

$$\epsilon = y_{LR,0} \delta$$

11.3 Volume of an Ideal Gas and Concentration

1 mol of an ideal gas at 0°C and 1 atm occupies $22.4 \text{ dm}^3 \left(0.0446 \frac{\text{mol}}{\text{dm}^3} \right)$.

1 lbmol of an ideal gas at 32°F and 1 atm occupies $359 \text{ ft}^3 \left(0.00279 \frac{\text{lbmol}}{\text{ft}^3} \right)$.

12 Heat Exchange Scenarios

12.1 Adiabatic

For adiabatic reactors, the relationship between conversion and temperature is the same for CSTRs, PFRs, PBRs, and batch reactors because the energy balance depends only on the initial and final thermodynamic states and not on the reactor path.

Conversion in terms of temperature:

$$X_{\text{EB}} = \frac{\sum_j \Theta_j C_{Pj} (T - T_0)}{-\Delta H_{rxn}^\circ (T_{ref}) - \Delta C_P (T - T_{ref})} \quad (64)$$

where ΔH_{rxn} should be written with respect to the limiting reactant.

Temperature in terms of conversion:

$$T = \frac{(-\Delta H_{rxn}^\circ (T_{ref})) X_{\text{EB}} + \sum_j \Theta_j C_{Pj} T_0 + X_{\text{EB}} \Delta C_P T_{ref}}{\sum_j \Theta_j C_{Pj} + X_{\text{EB}} \Delta C_P} \quad (65)$$

where ΔH_{rxn} should be written with respect to the limiting reactant. For an exothermic reaction, $\Delta H_{rxn} < 0$, so increasing conversion generally causes temperature to increase.

12.2 Constant T_a

Constant T_a assumes that the temperature of the cooling or heating fluid does not change significantly as it passes through the reactor. If this assumption can be made, you can use the equations in Section 7.

12.3 T_a is Not Constant

12.3.1 CSTR

Use the heat transfer balance for a perfectly mixed vessel:

$$\dot{Q} = \dot{m}_c C_{Pc} (T_{a0} - T) \left[1 - \exp \left(-\frac{UA}{\dot{m}_c C_{Pc}} \right) \right] \quad (66)$$

This expression comes from an energy balance on a perfectly mixed vessel coupled with a plug flow heat exchanger. $\dot{m}_c$ and C_{Pc} are the coolant mass flowrate and coolant heat capacity, respectively.

12.3.2 Tubular Reactors (PFR, PBR)

For a co-current flow:

$$\frac{dT_a}{dV} = \frac{Ua(T - T_a)}{\dot{m}_c C_{Pc}} \quad (67)$$

where $T_a = T_{a0}$ at $V = 0$.

For counter-current flow:

$$\frac{dT_a}{dV} = \frac{Ua(T_a - T)}{\dot{m}_c C_{Pc}} \quad (68)$$

where $T_a = T_{a0}$ at $V = V_{final}$. Since the coolant enters at the opposite end of the reactor, the coolant inlet condition occurs at the reactor outlet location. A guess-and-check procedure is typically required because the coolant boundary condition is specified at the opposite end of the reactor from the reactor boundary condition.

12.4 Batch Reactor

A transient energy balance must be written on the cooling fluid and coupled with the batch reactor mole and energy balance equations. Because both the reactor temperature and coolant temperature vary with time, the reactor and coolant energy balances must be solved simultaneously.

Quick Guide

- **Adiabatic:** Use Equation 64 or Equation 65.
- **Constant T_a :** Use the reactor energy balances in Section 7.
- **Variable T_a :** Add an energy balance for the cooling/heating fluid and solve it simultaneously with the reactor mole and energy balances.

13 Residence Time Distribution

13.1 Tracer Experiment Analysis

The residence time distribution is

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)dt} \quad (69)$$

where $C(t)$ is concentration vs. time data taken during pulse tracer experiment. It is typically integrated numerically, although regression and interpolation are sometimes used.

$E(t)$ is the residence time probability density function. It has units of time^{-1} . The quantity $E(t)dt$ represents the fraction of fluid with residence times between t and $t + dt$.

The cumulative residence time distribution is

$$F(t) = \int_0^t E(t') dt' \quad (70)$$

$F(t)$ represents the fraction of fluid that has exited the reactor by time t . It denotes the cumulative RTD and is unrelated to molar flowrate F .

$0 \leq F(t) \leq 1$, and $F(\infty) = 1$ (i.e., the sum of all probabilities = 1).

The mean residence time t_m is

$$t_m = \int_0^\infty t E(t) dt \quad (71)$$

This is analogous to the expected value (mean) of the probability distribution.

The residence time in an ideal PFR is

$$E(t) = \delta(t - \tau) \quad (72)$$

which is the Dirac delta function. In other words, all fluid elements in an ideal PFR spend exactly the same amount of time in the reactor.

The residence time in an ideal CSTR is

$$E(t) = \frac{\exp\left(-\frac{t}{\tau}\right)}{\tau} \quad (73)$$

The ideal CSTR residence time distribution reflects the phenomenon that some fluid elements leave the reactor almost as soon as they enter, while others remain much longer than the mean residence time. The ideal CSTR is a model; most real reactors exhibit different residence time distributions.

The dimensionless residence time distribution in an ideal CSTR is

$$E(\Theta) = \tau E(t) = \exp(-\Theta) \quad (74)$$

where $\Theta = \frac{t}{\tau}$. The advantage to using $E(\Theta)$ is that all ideal CSTR residence time distributions collapse onto a single curve.

13.2 Models for Estimating Reactor Conversion

For an ideal CSTR or PFR, $t_m = \tau$.

Non-ideal reactors can be modeled in a variety of ways. In this class, we use two extremes of mixing:

The *Segregation Model* assumes that fluid elements only “see” other fluid elements with the same residence time and do not mix with fluid elements with different residence times. In this model:

$$X_{seg} = \int_0^\infty X(t)E(t)dt \quad (75)$$

where $X(t)$ is obtained using the constant pressure batch reactor mole balance in coordination with the reaction rate law.

The *Maximum Mixedness* model assumes that the fluid is mixed at the molecular level, meaning that all fluid elements “see” all other fluid elements. This model is similar to perfect mixing, with the difference being the residence time distribution. In the maximum mixedness model:

$$\frac{dX}{dz} = \frac{-r_A}{C_{A0}} - \frac{E(\bar{T} - z)}{1 - F(\bar{T} - z)}X \quad (76)$$

This equation is provided for reference. It applies to the maximum-mixedness model for a single reaction under the assumptions used in

the derivation. In practice, it is typically solved numerically using interpolation of the RTD data.

In Equation 76, z is the time variable, $\bar{T}$ is the maximum time for which $C(t)$ data exists, and A is the limiting reactant. Equation 76 integrates backwards from $z = \bar{T}$ to $z = 0$ with the boundary condition that $X(\bar{T}) = 0$.

Quick Guide

Ideal PFR:

$$t_m = \tau, \quad E(t) = \delta(t - \tau)$$

Ideal CSTR:

$$t_m = \tau, \quad E(t) = \frac{1}{\tau} \exp(-t/\tau)$$

Real reactors: Use measured $E(t)$ with the Segregation or Maximum Mixedness model.

14 Numerical Methods Cheat Sheet

- **Interpolation:** Estimate values between data points. Common uses in CBE 3610 include RTD analysis and evaluating experimentally measured properties.
- **Regression:** Fit a mathematical model to data. Common uses include determining reaction orders, rate constants, and kinetic parameters from experimental data.
- **Numerical Integration:** Evaluate definite integrals numerically. Common uses include Levenspiel plots, RTD calculations, and reactor-design equations that cannot be integrated analytically.

- **Root Finding:** Solve equations of the form

$$f(x) = 0.$$

Common uses include solving algebraic mole balances, equilibrium relationships, and energy balances for conversion or temperature.

- **ODE Solver:** Solve ordinary differential equations such as

$$\frac{dX}{dt}, \quad \frac{dF_A}{dV}, \quad \frac{dT}{dV}.$$

Common uses include PFR, PBR, and batch reactor design problems.

15 Other Useful Equations

The equations in this section are used less frequently than the reactor design equations presented earlier but are useful for problems involving nonideal flow, pressure drop, reaction mechanisms, and diffusion limitations.

15.0.1 General Balance Equations

General mole balance:

$$F_{j,in} - F_{j,out} + \int_V r_j dV = \frac{dN_j}{dt} \quad (77)$$

General energy balance, neglecting kinetic and potential energy:

$$\frac{d\hat{E}_{sys}}{dt} = \dot{Q} - \dot{W}_s + \sum_{i=1}^n F_{i0} H_{i0} - \sum_{i=1}^n F_i H_i \quad (78)$$

15.0.2 Accounting for Pressure Drop

Ergun equation (full and general equation):

$$\frac{dP}{dz} = -\frac{G}{\rho g_c D_P} \left(\frac{1-\phi}{\phi^3} \right) \left[\frac{150(1-\phi)\mu}{D_P} + 1.75G \right] \quad (79)$$

Simplified Ergun equation (collapses some parameters into a single parameter, general equation):

$$\frac{dp}{dW} = -\frac{\alpha}{2p} \frac{T}{T_0} \frac{F_T}{F_{T0}} \quad (80)$$

where p is the reduced pressure equal to P_T/P_{T0} and α is a lumped pressure drop parameter that combines particle, flow, and bed properties.

The simplified Ergun Equation with respect to conversion, X :

$$\frac{dp}{dW} = -\frac{\alpha}{2p} (1 + \epsilon X) \frac{T}{T_0} \quad (81)$$

15.0.3 Additional Models for Non-Ideal Reactors

The Axial Dispersion Model:

$$D_{ax} \frac{d^2 C_A}{dz^2} - u \frac{dC_A}{dz} + r_A = 0 \quad (82)$$

represents deviations from plug flow due to axial mixing (backmixing). Although D_{ax} has the same units as molecular diffusivity (D_{AB}) and effective diffusivity (D_e), it is usually much larger than these quantities because it represents the combined effects of velocity variations, turbulence, channeling, and other hydrodynamic mixing phenomena that cause departures from ideal plug flow.

The Laminar Flow Reactor Model: Velocity varies across the reactor cross section, causing different fluid elements to experience different residence times. Whereas the Axial Dispersion Model represents non-ideality through axial mixing, the Laminar Flow Reactor Model represents non-ideality through the velocity profile. Analysis is generally performed using residence-time- distribution methods or by integrating over the velocity profile.

15.0.4 Pseudo-Steady-State Hypothesis (PSSH)

Pseudo-Steady-State Hypothesis (PSSH):

$$r_{A^*} = \sum_{i=1}^n r_{iA^*} = 0 \quad (83)$$

where A^* is an active intermediate. The PSSH assumes that active intermediates do not accumulate, i.e., that $\frac{dC_{A^*}}{dt} = 0$.

15.0.5 Diffusion Limited Catalyzed Reactions

Thiele modulus for diffusion limited catalytic systems, ϕ_n , where n is the reaction order, compares characteristic rates of reaction and diffusion within a catalyst particle.

$$\phi_n = R \sqrt{\frac{k_n C_{As}^{n-1}}{D_e}} \quad (84)$$

where s indicates conditions at the pellet surface.

Internal effectiveness factor for diffusion limited catalytic systems

$$\eta = \frac{\text{actual rate of overall reaction}}{\text{rate of reaction that would result if } C_A \text{ inside the pellet} = C_{As}} \quad (85)$$

For large values of ϕ_n :

$$\eta = \sqrt{\frac{2}{n+1}} \frac{3}{\phi_n} \quad (86)$$

This approximation is valid for large values of ϕ_n .

16 Unit Conversion Factors

16.1 Volume

$$\begin{array}{ll} 1 \text{ cm}^3 = 0.001 \text{ dm}^3 & 1 \text{ in}^3 = 0.0164 \text{ dm}^3 \\ 1 \text{ fluid oz} = 0.0296 \text{ dm}^3 & 1 \text{ ft}^3 = 28.32 \text{ dm}^3 \\ 1 \text{ m}^3 = 1000 \text{ dm}^3 & 1 \text{ U.S. gallon} = 3.785 \text{ dm}^3 \\ 1 \text{ L} = 1 \text{ dm}^3 & 1 \text{ ft}^3 = 7.482 \text{ gal} \end{array}$$

16.2 Length

$$\begin{array}{ll} 1 \text{ \AA} = 10^{-8} \text{ cm} & 1 \text{ dm} = 10 \text{ cm} \\ 1 \text{ }\mu\text{m} = 10^{-4} \text{ cm} & 1 \text{ in} = 2.54 \text{ cm} \\ 1 \text{ ft} = 30.48 \text{ cm} & 1 \text{ m} = 100 \text{ cm} \end{array}$$

16.3 Pressure

$$\begin{array}{ll} 1 \text{ torr} = 0.13333 \text{ kPa} & 1 \text{ torr} = 1 \text{ mmHg} \\ 1 \text{ in. H}_2\text{O} = 0.24886 \text{ kPa} & 1 \text{ Pa} = 1 \frac{\text{N}}{\text{m}^2} \\ 1 \text{ in. Hg} = 3.3843 \text{ kPa} & 1 \text{ atm} = 101.33 \text{ kPa} \\ 1 \text{ psia} = 6.8943 \text{ kPa} & 1 \frac{\text{megadyne}}{\text{cm}^2} = 100 \text{ kPa} \end{array}$$

16.4 Temperature

$$^{\circ}\text{F} = 1.8 \times ^{\circ}\text{C} + 32 \quad ^{\circ}\text{R} = ^{\circ}\text{F} + 459.69$$

$$\text{K} = ^{\circ}\text{C} + 273.15 \quad ^{\circ}\text{R} = 1.8 \times \text{K}$$

16.5 Mass

$$1 \text{ lb}_m = 454 \text{ g} \quad 1 \text{ grain} = 0.0648 \text{ g}$$

$$1 \text{ oz} = 28.35 \text{ g} \quad 1 \text{ U.S. short ton} = 908,000 \text{ g}$$

16.6 Energy

$$1 \text{ Btu} = 1055.06 \text{ J} \quad 1 \text{ cal} = 4.1868 \text{ J}$$

$$1 \text{ L} \cdot \text{atm} = 101.34 \text{ J} \quad 1 \text{ hp} \cdot \text{h} = 2.6806 \times 10^6 \text{ J}$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J} \quad 1 \text{ kg} \cdot \text{m}^2/\text{s}^2 = 1 \text{ J}$$

16.7 Viscosity

$$1 \text{ poise} = 1 \frac{\text{g}}{\text{cm} \cdot \text{s}} \quad 1 \text{ centipoise} (1 \text{ cp}) = 1 \text{ mPa} \cdot \text{s} = 10^{-3} \text{Pa} \cdot \text{s}$$

16.8 Force

$$1 \text{ dyne} = 1 \frac{\text{g} \cdot \text{cm}}{\text{s}^2} \quad 1 \text{ Newton} = 1 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$1 \text{ lb}_f = 4.448 \text{ N} \quad 1 \text{ lb}_f = 32.174 \frac{\text{ft} \cdot \text{lb}_m}{\text{s}^2}$$

16.9 Power

$$1 \text{ Watt} (1 \text{ W}) = 1 \frac{\text{J}}{\text{s}} \quad 1 \text{ hp} = 746 \text{ W}$$

16.10 Work

Work = Force $\times$ Distance

Work = $\int PdV$. For constant P, $W = P \times V$.

16.11 Gravity and the Gravitational Conversion Factor (g_c)

Gravitational constant: $g = 32.174 \frac{\text{ft}}{\text{s}^2}$

Gravitational conversion factor: $g_c = \frac{32.174 \text{ ft lb}_m}{\text{s}^2 \text{ lb}_f}$

g_c in SI units is equal to 1.

17 Useful Definite Integrals

$$\int_0^x \frac{dX}{1-X} = \ln \left(\frac{1}{1-x} \right) \quad (87)$$

$$\int_{x_1}^{x_2} \frac{dX}{(1-X)^2} = \frac{1}{1-x_2} - \frac{1}{1-x_1} \quad (88)$$

$$\int_0^x \frac{dX}{(1-X)^2} = \frac{x}{1-x} \quad (89)$$

$$\int_0^x \frac{dX}{1+\epsilon X} = \frac{1}{\epsilon} \ln(1+\epsilon x) \quad (90)$$

$$\int_0^x \frac{(1 + \epsilon X) dX}{1 - X} = (1 + \epsilon) \ln \left(\frac{1}{1 - x} \right) - \epsilon x \quad (91)$$

$$\int_0^x \frac{(1 + \epsilon X) dX}{(1 - X)^2} = \frac{(1 + \epsilon) x}{1 - x} - \epsilon \ln \left(\frac{1}{1 - x} \right) \quad (92)$$

$$\int_0^x \frac{(1 + \epsilon X)^2 dX}{(1 - X)^2} = 2\epsilon(1 + \epsilon) \ln(1 - x) + \epsilon^2 x + \frac{(1 + \epsilon)^2 x}{1 - x} \quad (93)$$

$$\int_0^x \frac{dX}{(1 - X)(\Theta_B - x)} = \frac{1}{\Theta_B - 1} \ln \left(\frac{\Theta_B - x}{\Theta_B(1 - x)} \right) \quad (94)$$

where $\Theta_B \neq 1$.

$$\int_0^w (1 - \alpha W)^{\frac{1}{2}} dW = \frac{2}{3\alpha} \left[1 - (1 - \alpha w)^{\frac{3}{2}} \right] \quad (95)$$

$$\int_0^x \frac{dX}{aX^2 + bX + c} = \frac{-2}{2ax + b} + \frac{2}{b} \quad (96)$$

where $b^2 = 4ac$.

$$\int_0^x \frac{dX}{aX^2 + bX + c} = \frac{1}{a(p - q)} \ln \left(\frac{q x - p}{p x - q} \right) \quad (97)$$

for $b^2 > 4ac$, where p and q are the roots of the equation $aX^2 + bX + c$, i.e., $p, q = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$\int_0^x \frac{a + bX}{c + gX} dX = \frac{bx}{g} + \frac{ag - bc}{g^2} \ln \left(\frac{c + gx}{c} \right) \quad (98)$$

18 Symbol Definitions

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Symbol	Example units
A = any reactive species	-
A = area	m^2
A = pre-exponential (frequency) factor	same as k
A = Helmholtz free energy	kJ/mol
a = tubular reactor heat exchange area	m^2/m^3
B = any reactive species	-
C = concentration	mol/L
C_P = heat capacity	$\frac{\text{J}}{\text{mol}\cdot\text{K}}$
C = any reactive species	-
D_P = diameter of catalyst particle	m
D_e = effective diffusivity	m^2/s
D_{ax} = axial diffusion coefficient	m^2/s
D = any reactive species	-
D = desired product	-
E = activation energy	kJ/mol
ΔE^{act} = activation energy	kJ/mol
E = system energy	kJ/mol
$E(t)$ = residence time distribution function	min^{-1}

F = molar flowrate	mol/s
F_T = total molar flowrate	mol/s
$F(t)$ = cumulative residence time distribution	unitless
F = Helmholtz free energy	kJ/mol
g_c = gravitational conversion factor	$\frac{\text{lb}_m \cdot \text{ft}}{\text{s}^2 \cdot \text{lb}_f}$
G = Gibbs free energy	kJ/mol
G = superficial mass velocity = ρu	$\frac{\text{kg}}{\text{m}^2 \cdot \text{s}}$
$G(T)$ = heat generated	kJ/mol
H = enthalpy	kJ/mol
ΔH_{rxn} = heat of reaction	kJ/mol
K_p = pressure equilibrium constant	atm^δ
K_c = concentration equilibrium constant	$\frac{\text{mol}^\delta}{\text{L}^\delta}$
k = rate constant	$\frac{\text{L}^n}{\text{mol}^n \cdot \text{s}}, n = \text{reaction order} - 1$
LR = limiting reactant	-
$\dot{m}_c$ = mass flowrate of coolant	kg/s
N = number of moles	mol
P = pressure	atm
p = reduced pressure = $\frac{P_T}{P_{T0}}$	unitless
p = momentum	$\text{N} \cdot \text{s}$
Q = heat	kJ
Q = reaction quotient	unitless

Q_g = heat generated	kJ/s/L
Q_r = heat removed	kJ/s/L
R = gas constant	$\frac{\text{J}}{\text{mol}\cdot\text{K}}$ or $\frac{\text{L}\cdot\text{atm}}{\text{mol}\cdot\text{K}}$
R = radius of a catalyst pellet	m
$R(T)$ = heat removed	kJ/mol
r = rate	$\frac{\text{mol}}{\text{dm}^3\cdot\text{s}}$
r' = rate	$\frac{\text{mol}}{\text{kg catalyst}\cdot\text{s}}$
S = entropy	$\frac{\text{J}}{\text{mol}\cdot\text{K}}$
S = selectivity	moles of D/moles of U
T = temperature	K
T_a = ambient or cooling fluid temperature	K
T_R = reaction temperature	K
T_{ref} = reference temperature, usually 298 K	K
$\bar{T}$ = maximum time	min
t = time	s
t_m = mean residence time	min
U = overall heat transfer coefficient	$\frac{\text{J}}{\text{m}^2\cdot\text{s}\cdot\text{K}}$
Ua = volumetric heat transfer coefficient ($= U \frac{dA}{dV}$)	$\frac{\text{J}}{\text{m}^3\cdot\text{s}\cdot\text{K}}$
U = velocity	m/s
U = internal energy	kJ/mol
U = undesired product	-

u = velocity	m/s
V = volume	dm ³
v = volumetric flowrate	dm ³ /min
v = velocity	m/s
W = weight of catalyst	kg
W_s = shaft work	kJ/mol
X = conversion	unitless
X_e = equilibrium conversion	unitless
x = mole fraction (liquid phase)	moles of A per mole
Y = yield	moles of D/moles of A
y = mole fraction (vapor phase)	moles of A per mole
z = reactor length	m
z = time	min
α = pressure drop parameter	kg ⁻¹
β = approach to equilibrium	unitless
δ = Mole change parameter	moles per mole of A
δ = Dirac delta	-
ϵ = Mole change parameter	unitless
Θ = ratio of moles or molar flowrates	moles of B per mole of LR
Θ = dimensionless time	unitless
μ = viscosity	$\frac{\text{kg}}{\text{m}\cdot\text{s}}$
ν = stoichiometric coefficient	unitless

ρ = density	kg/m ³
ρ_b = bulk density of catalyst	$\frac{\text{kg catalyst}}{\text{L of reactor bed}}$
τ = residence time	s
ϕ = void fraction of packed bed	unitless
ϕ_n = Thiele modulus for an n order reaction	unitless
ξ = extent of reaction	mol